

Introduction to Measurement Models

SMIP Summer School 2025: Lecture 02

Section Objectives

1. Latent traits
2. Our first graphical model (path diagram)
3. Psychometric models from generalized linear models

Latent Traits: A Big-Picture View

Latent trait theory posits there are attributes of a person (typically) that are:

- Unobservable (hence the term latent)
- Quantifiable
- Related to tasks that can be observed

Often, these attributes are often called constructs, underscoring they are constructed and do not exist, such as:

- A general or specific ability (in educational contexts)
- A feature of personality, such as “extroversion” (in psychological contexts)

The same psychometric models apply regardless of the measurement context

Latent Traits are Interdisciplinary

- Many varying fields use some version of latent traits
- Similar (or identical) methods are often developed separately
 - Item response theory in education
 - Item factor analysis in psychology
- Many different terms for same ideas, such as the
 - Label given to the latent trait: Factor/Ability/Attribute/Construct/Variable/True Score/Theta (θ)
 - Label given to those giving the data: Examinee/Subject/Participant/Respondent/Patient/Student
 - Label given to the data: Outcome/Indicator/Item/Response
- What this means:
 - Lots of words to keep track of, but (relatively) few concepts

Best Measures are Built Purposefully

- Latent constructs seldom occur randomly—they are defined
 - The definition typically indicates:
 - What the construct means
 - What observable behaviors are likely related to the construct
 - For a lot of what we do, observable behavior means answering questions on an assessment or survey
- Therefore, modern psychometric methods are built around specifying the set of observed variables to which a latent variable relates
 - No need for exploratory analyses—we define our construct and seek to falsify our definition
- The term I use for “relates” is “measure”
 - e.g., Educational assessment items measure some type of ability

Guiding Principles

- To better understand psychometric methods and theory, envision what analyses would be used if latent variables were not latent
 - Example: Imagine if we could directly observe mathematics ability
- Then, consider what we would do with that value
 - Example: We could predict how students would perform on items using logistic regression (with θ as an observed predictor)
- Psychometric models essentially do this—use observed variable methods as if we know the value of the latent variable
 - But, with:
 - A data collection design allowing for such methods to be used
 - A more formal vetting of whether or not we did a good job measuring the construct

Measurement of Latent Constructs

How does this process differ when we cannot observe the thing we are measuring—when the construct is latent?

- We still need something we can observe—item responses for example
- We need a method to map the response to a number (Strongly agree==5?)
- We also need a way to aggregate all responses to a value that represents a person
 - A score or classification
- We then need a way to ensure what we just did means what we think it does
 - Methods for validation
- We also need to remember that the values we estimate for a person's latent trait(s) won't be perfectly reliable
 - Caution needed for secondary analyses

Measurement Models

- Distinguishing feature of psychometric models is the second word—they are models
 - We often call such models “measurement models”
- Measurement models are the mathematical specification that provides the link between the latent variable(s) and the observed data
- The form of such models looks different across the wide classes of measurement models (e.g., factor analysis vs. item response models) but wide generalities exist
 - Continuous Responses? “Confirmatory Factor Analysis”
 - Categorical responses? “Item Response Theory or Item Factor Analysis”
- Measurement models need:
 - Distributional assumptions about the data (with link functions)
 - A linear or non-linear form that predicts data from the trait(s)
- The key: Observed data are being predicted by latent variable(s)

Models vs. Other Techniques

Measurement models are a different way of thinking about psychometrics than what most people without psychometric training do

- Most scientists enumerate item response scores (e.g., correct response == 1; strongly agree == 5)
- The latent trait score estimate is then formed by summing the response scores
 - “Add ‘Stuff’ Up” model: (Scheiße zusammenzählen?)
- As it turns out, the naive adding together of item scores implies a measurement model
 - Called parallel items — very strict assumptions
 - Equal variances and covariances for all observed variables
 - Items measure the trait equally well, with the same amount of error, and no additional relations

Characteristics of Latent Variables

Latent variables can be defined to have different levels of measurement

- Interval level (as in factor analysis and item response theory) — Continuous
 - No absolute zero; units of the factor are equivalent across the range of values
 - Example: A person with a value of 2 is the same distance from a person with a value of 0 as is a person with a value of -2
- Ordinal level (as in diagnostic classification models)
 - Can rank order people but not determine how far apart they may be
 - Example: Students considered masters of a topic have more ability than students considered non-masters
- Nominal level (as in latent class or finite mixture models) — Categorical
 - Groups/classes/clusters of people

Most Common: Continuous Latents

- For most of this workshop, we will treat latent variables as continuous (interval level)
- As they do not exist, continuous latent variables need a defined metric:
 - What is their mean? (Usually set to zero)
 - What is their standard deviation? (Set to one *or* borrowed from metric of items)
- Defining the metric is the first step in a latent variable model
 - Called scale identification
- The metric is arbitrary
 - Can set differing means/variances but still have same model
 - Linear transformations of parameters based on scale mean and standard deviation

Measurement Model Path Diagrams

Measurement models are often depicted in graphical format, using what is called a path diagram

- Typically, latent variables are represented as objects that are circles/ovals
- Using graph theory terms, a variable in a path diagram (latent or observed) is called a node
- Lines connecting the variables are called edges

Latent Variable Only



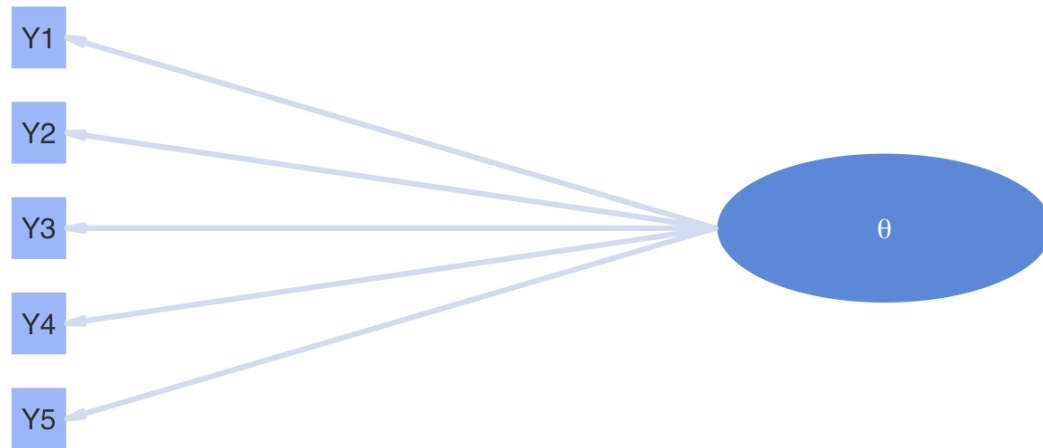
Adding Observed Variables

Measurement model path diagrams often denote observed variables with rectangular boxes

On the next slide:

- The term “latent variable” is replaced with θ
- The observed variables are denoted as $Y1$ through $Y5$
 - Imagine these represent five observed items of a scale measuring θ

Path Diagram with Observed and Latent Variables



Path Diagrams: Not Models

Path diagrams are useful for depicting a measurement model but are not isomorphic with the mathematical models they depict

- All model parameters are often not included in the diagram
- No indication about the distribution of the variables

Translating a Path Diagram to a Model

Going back to the point from before—let's imagine the latent variable as an observed variable

- An arrow (edge) indicates one variable predicts another
 - The predictor is the variable on the side of the arrow without the point
 - The outcome is the variable on the side of the point
- If we assume the items were continuous (like linear regression), the diagram indicates a regression model for each outcome

$$Y_{p1} = \beta_{Y_1,0} + \beta_{Y_1,1}\theta_p + e_{p,Y_1}$$

Interpreting the Parameters

All five regression lines implied by the model are then:

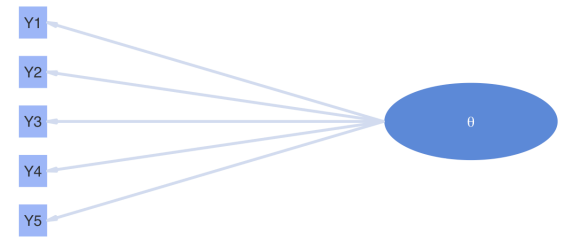
$$Y_{p1} = \beta_{Y_1,0} + \beta_{Y_1,1}\theta_p + e_{p,Y_1}$$

$$Y_{p2} = \beta_{Y_2,0} + \beta_{Y_2,1}\theta_p + e_{p,Y_2}$$

$$Y_{p3} = \beta_{Y_3,0} + \beta_{Y_3,1}\theta_p + e_{p,Y_3}$$

$$Y_{p4} = \beta_{Y_4,0} + \beta_{Y_4,1}\theta_p + e_{p,Y_4}$$

$$Y_{p5} = \beta_{Y_5,0} + \beta_{Y_5,1}\theta_p + e_{p,Y_5}$$



- $\beta_{Y_i,0}$ is the intercept of the regression line predicting the score from item Y_i
 - The expected response score for a person who has $\theta_p = 0$
- $\beta_{Y_i,1}$ is the slope of the regression line predicting the score from item Y_i
 - The expected change in the response score for a one-unit change in θ_p

More Interpreting the Parameters

Also:

- e_{p,Y_i} is the residual (error), indicating the difference in the predicted score for person p to item i
 - Like in regression, we additionally assume:
 - $e_{p,Y_i} \sim N(0, \sigma_{e_{Y_i}}^2)$: is normally distributed with mean zero
- $\sigma_{e_{Y_i}}^2$ is the residual variance of item Y_i , indicating the square of how far off the prediction is on average

The five regression models are estimated simultaneously:

- If θ_p were observed, we would call this a multivariate regression
 - Multivariate regression: Multiple continuous outcomes predicted by one or more predictors

More About Regression

$$Y_{pi} = \beta_{Y_i,0} + \beta_{Y_i,1}\theta_p + e_{p,Y_i}$$

In the regression model for a single variable, what distribution do we assume about the outcome?

- As error is normally distributed, the outcome takes a normal distribution
$$Y_{pi} \sim N(?, ?)$$
- As $\beta_{Y_i,0}$, $\beta_{Y_i,1}$, and θ_p are constants, they move the mean of the outcome to $\beta_{Y_i,0} + \beta_{Y_i,1}\theta_p$
 - $Y_{pi} \sim N(\beta_{Y_i,0} + \beta_{Y_i,1}\theta_p, ?)$
- As error has a variance of $\sigma_{e_{Y_i}}^2$, the outcome is assumed to have variance $\sigma_{e_{Y_i}}^2$
 - $Y_{pi} \sim N(\beta_{Y_i,0} + \beta_{Y_i,1}\theta_p, \sigma_{e_{Y_i}}^2)$
- Therefore, we say Y_{pi} follows a conditionally normal distribution

The Univariate Normal Distribution

$Y \sim N(\mu, \sigma^2)$ that implies a probability density function (pdf)

$$f(Y) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(Y - \mu)^2}{2\sigma^2}\right]$$

- Here, π is the constant 3.14 and $\exp$ is Euler's constant (2.71)
- Of note here is that there are three components that go into the function:
 - The data Y
 - The mean μ — this can be the conditional mean we had on the previous slide (formed by parameters)
 - The variance σ^2
- The key to using Bayesian methods is to know the distributions for each of the variables in the model

From Regression to CFA

When θ_i is latent, the five-variable model becomes a confirmatory factor analysis (CFA) model

- CFA: Prediction of continuous items using linear regression with one or more continuous latent variables as predictors
 - The interpretations of the regression parameters are identical between linear regression and CFA

Regression and CFA Differences

The differences between CFA and regression are:

- θ_p as a predictor is not observed in CFA but is observed in regression
 - Therefore, we must set its mean and variance
 - There are multiple ways to do this (standardized factor, marker item, etc...)—stay tuned
- Each of the model parameters has a different name (and symbol denoting it) in CFA
 - $\beta_{Y_i,0} = \mu_i$ is the item intercept
 - $\beta_{Y_i,1} = \lambda_i$ is the factor loading for an item
 - $\sigma_{e_{Y_i}}^2 = \psi_i^2$ is the unique variance for an item
- Need a sufficient number of observed variables to empirically identify the latent trait
 - Three items for identification (and four for testable unidimensionality)

Changing Notation

Our five-item CFA model with CFA-notation:

$$Y_{p1} = \mu_1 + \lambda_1 \theta_p + e_{p,Y_1}$$

$$Y_{p2} = \mu_2 + \lambda_2 \theta_p + e_{p,Y_2}$$

$$Y_{p3} = \mu_3 + \lambda_3 \theta_p + e_{p,Y_3}$$

$$Y_{p4} = \mu_4 + \lambda_4 \theta_p + e_{p,Y_4}$$

$$Y_{p5} = \mu_5 + \lambda_5 \theta_p + e_{p,Y_5}$$

Measurement Models for Different Item Types

- The CFA model assumes (1) continuous latent variables and (2) continuous item scores
 - What should we do if we have binary items (e.g., yes/no, correct/incorrect)?
- If we had observed θ_p and wanted to predict $Y_{1p} \in \{0, 1\}$ what type of analysis would we use?
- Logistic regression:

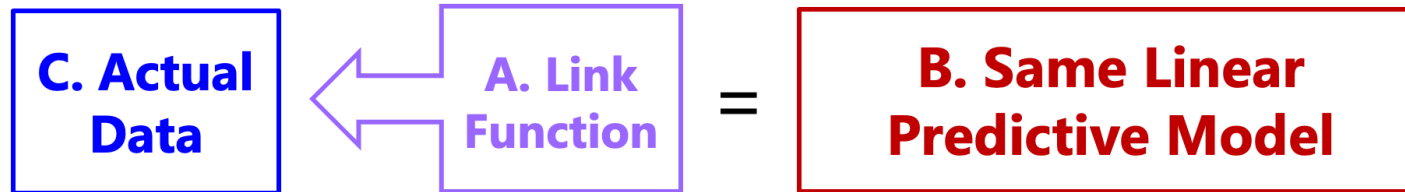
$$P(Y_{p1} = 1) = \frac{\exp(\beta_{Y_1,0} + \beta_{Y_1,1}\theta_p)}{1 + \exp(\beta_{Y_1,0} + \beta_{Y_1,1}\theta_p)}$$

From CFA to IRT and IFA

Outcome Type → <i>Model Family Name</i>	Observed Predictor x_i	Latent Predictor x_i
Continuous outcomes → <i>"General Linear Model"</i>	"Linear" Regression	Confirmatory Factor Models
Discrete/categorical outcomes → <i>"Generalized Linear Model"</i>	Logistic/Probit/ Multinomial Regression	Item Response Theory and Item Factor Analysis

- Basis of Item Response Theory (IRT) and Item Factor Analysis (IFA) lies in models for discrete outcomes, which are called "**generalized**" linear models
- Thus, IRT and IFA will be easier to understand after reviewing concepts from *generalized* linear models...

3 Parts of Generalized Linear Models



- A. Link Function: Transformation of *conditional mean* to keep predicted outcomes within the bounds of the outcome
- B. Same Linear Model: How the model linearly predicts the *link-transformed* conditional mean of the outcome
- C. Conditional Distribution: How the outcome could be distributed given the possible values of the outcome

Generalized linear models work for many kinds of outcomes...

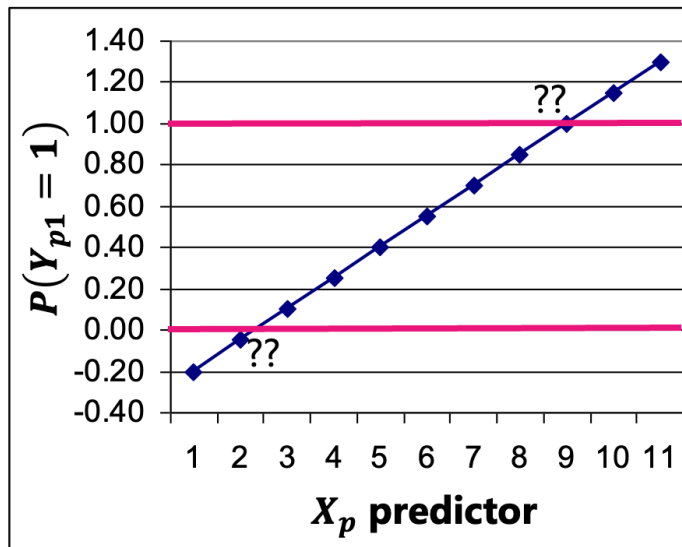
Here's how it works for binary outcomes

- Let's say we have a single binary (0 or 1) outcome... Y_{p1}
- The mean of a binary outcome is the proportion of 1 values
 - So given each person's predictor values, the model tries to predict the **conditional mean**: the **probability of a 1**: $P(Y_{p1} = 1)$
 - The conditional mean has more possible values than the outcome!
 - General linear model: $P(Y_{p1} = 1) = \beta_{Y_1,0} + \beta_{Y_1,1}(X_p) + e_{p,Y_1}$
 - $\beta_{Y_1,0}$ = expected probability of $Y_{p1} = 1$ when all predictors = 0
 - $\beta_{Y_1,1}$ = expected change in $P(Y_{p1} = 1)$ for per unit change in X_p
 - e_{p,Y_1} = difference of observed minus predicted **binary** values
 - Model becomes $y_i = (\text{predicted probability of 1}) + e_{p,Y_1}$
 - **What could possibly go wrong???**

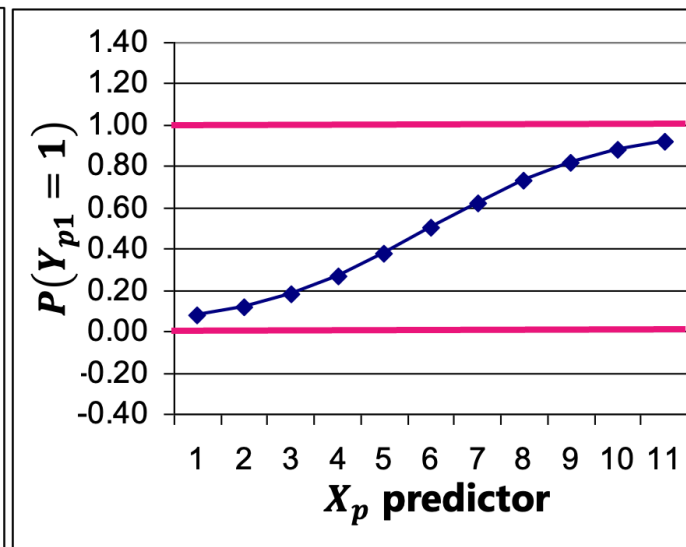
Normal GLM for Binary Outcomes?

- Problem #1: A **linear relationship** between X_p and Y_{p1} ???
- Probability of a 1 is bounded between 0 and 1, but predicted probabilities from a linear model aren't going to be bounded
- Linear relationship needs to shut off $\rightarrow$ made nonlinear

We have this...



But we need this...



Generalized Models for Binary Outcomes

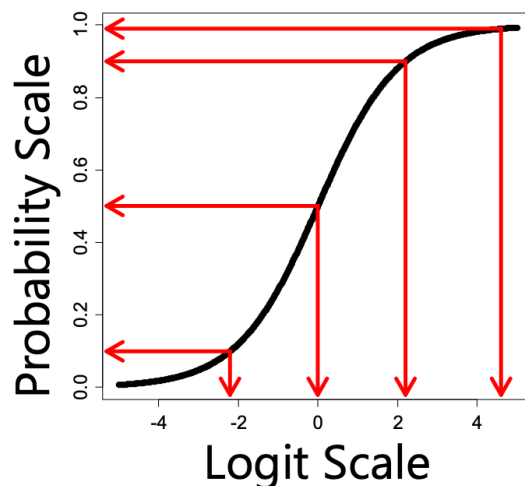
- Solution to #1: Rather than predicting $P(Y_{p1} = 1)$ directly, the model transforms it into an unbounded outcome using a **link function**:

➤ Transform **probability** into **odds**: $\frac{P(Y_{p1}=1)}{1-P(Y_{p1}=1)} = \frac{P(Y_{p1}=1)}{P(Y_{p1}=0)}$

- If $P(Y_{p1} = 1) = .7$ then Odds(1) = 2.33; Odds(0) = 0.429
- But odds scale is skewed, asymmetric, and ranges 0 to $+\infty \rightarrow$ Not a good outcome!

➤ Take **natural log of odds** $\rightarrow$ called “**logit**” link: $\log \left[\frac{P(Y_{p1}=1)}{1-P(Y_{p1}=1)} \right]$

- If $p(y_i = 1) = .7$, then Logit(1) = 0.846; Logit(0) = -0.846
- Logit scale is now symmetric about 0, range is $\pm\infty \rightarrow$ Now a good outcome to predict!

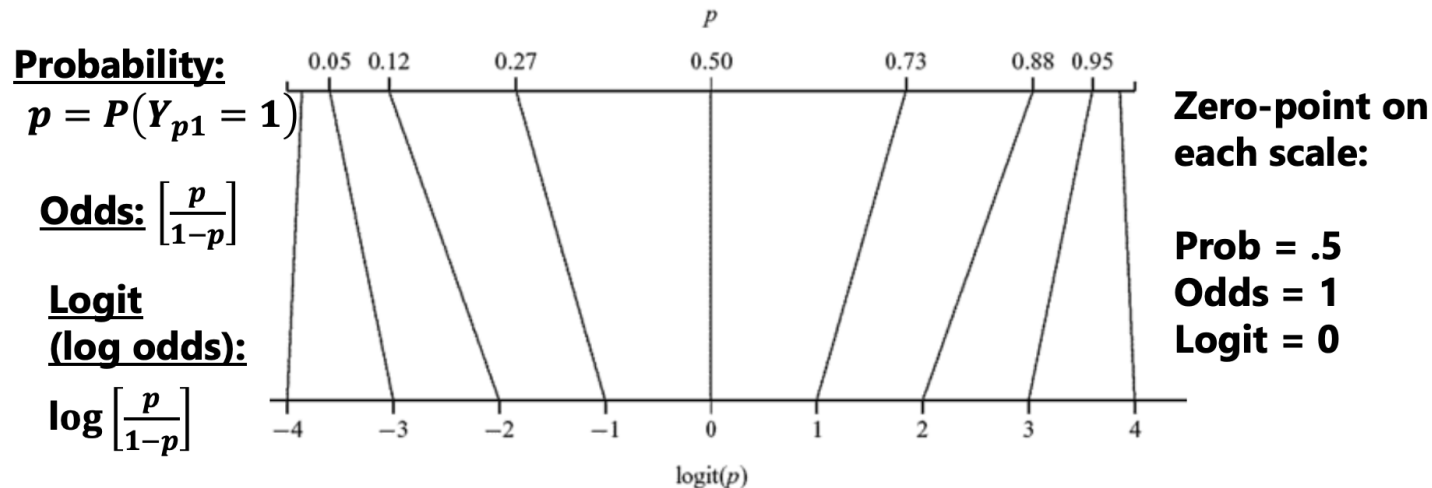


Probability $\rightarrow$ “data scale”	Logit $\rightarrow$ “model scale”
0.99	4.6
0.90	2.2
0.50	0.0
0.10	-2.2

Can you guess
what $P(.01)$
would be on
the logit scale?

Solution to #1: Probability to Logits

- A Logit link is a nonlinear transformation of probability:
 - Equal intervals in logits are NOT equal intervals of probability
 - Logits range from $\pm\infty$ and are symmetric about prob = .5 ($\rightarrow$ logit = 0)
 - Now we can use a linear model $\rightarrow$ The model will **linearly predict the expected logit**, which translates into a nonlinear prediction of probability
 $\rightarrow$ **the outcome conditional mean (probability) shuts off at 0 or 1 as needed**



Normal GLM for Binary Outcomes?

- General linear model: $P(Y_{p1} = 1) = \beta_{Y_1,0} + \beta_{Y_1,1}(X_p) + e_{p,Y_1}$
- But e_{p,Y_1} can only be 2 things given: $e_{p,Y_1} = Y_{p1} - \hat{Y}_{p1}$
 - If $Y_{p1} = 0$ then $e_{p,Y_1} = (0 - \text{predicted probability})$
 - If $Y_{p1} = 1$ then $e_{p,Y_1} = (1 - \text{predicted probability})$
- Problem #2a: So the residuals can't be normally distributed
- Problem #2b: The residual variance can't be constant over $\hat{Y}_{p1}$ as in GLM because the **mean and variance are dependent**
 - Variance of binary variable: $Var(Y_{p1}) = p * (1 - p)$

Mean and Variance of a Binary Variable

Mean (p)	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9	1.0
Variance	.0	.09	.16	.21	.24	.25	.24	.21	.16	.09	.0

Interpreting Model Parameters

$$\text{Logit} \left(P \left(Y_{p1} = 1 \right) \right) = \beta_{Y_1,0} + \beta_{Y_1,1} \theta_p$$

Here:

- $\beta_{Y_1,0}$ is the intercept — the expected log odds of a correct response when $\theta_p = 0$
- $\beta_{Y_1,1}$ is the slope — the expected change in log odds of a correct response for a one-unit change in θ_p
- Note: there is no error variance term

Bernoulli Distributions

- Using logistic regression for binary outcomes makes the assumption that the outcome follows a (conditional) Bernoulli distribution, or $Y \sim B(p_Y)$
 - The parameter p_Y is the probability that Y equals one, or $P(Y = 1)$
- The Bernoulli pdf (sometimes called the probability mass function as the variable is discrete) is:

$$f(Y) = (p_Y)^Y (1 - p_Y)^{1-Y}$$

- So, there is no error variance parameter in logistic regression as there is no parameter in the distribution that represents error (it is a non-constant function of the mean)
 - Error is represented by how far off a probability is from either zero or one

Logistic Regression with Latent Variable(s)

- Back to our running example, if we had binary items and wished to form a (unidimensional) latent variable model, we would have something that looked like:

$$P(Y_{pi} = 1 \mid \theta_p) = \frac{\exp(\mu_i + \lambda_i \theta_p)}{1 + \exp(\mu_i + \lambda_i \theta_p)}$$

Logistic Regression with Latent Variable(s)

$$P(Y_{pi} = 1 \mid \theta_p) = \frac{\exp(\mu_i + \lambda_i \theta_p)}{1 + \exp(\mu_i + \lambda_i \theta_p)}$$

- Here, the parameters retain their names from CFA:
 - $\beta_{Y_i,0} = \mu_i$ is the item intercept
 - $\beta_{Y_i,1} = \lambda_i$ is the factor loading for an item
- We call this slope-intercept parameterization
- This parameterization is called item factor analysis(IFA)
 - Sometimes the intercept μ_i is replaced with a threshold τ (where $\tau = -\mu_i$)

From IFA to IRT

- IFA and IRT are equivalent models—their parameters are transformations of each other:

$$a_i = \lambda_i$$
$$b_i = -\frac{\mu_i}{\lambda_i}$$

From IFA to IRT

IFA with "easiness" **intercept** μ_i : $\text{Logit}(Y_{pi} = 1) = \mu_i + \lambda_i \theta_p$ $\mu_i = -\tau_i$

IFA with "difficulty" **threshold** τ_i : $\text{Logit}(Y_{pi} = 1) = -\tau_i + \lambda_i \theta_p$

IFA model with "difficulty" thresholds can be written as a **2PL IRT Model**:

IRT model:

$$\text{logit}(Y_{pi} = 1) = a_i(\theta_s - b_i) = -\underbrace{a_i b_i}_{\tau_i} + \underbrace{a_i \theta_p}_{\lambda_i}$$

IFA model:

a_i = discrimination
 b_i = difficulty
 θ_p = latent trait

Convert IFA to IRT:

$$a_i = \lambda_i * \sqrt{\text{theta variance}}$$

$$b_i = \frac{\tau_i - (\lambda_i * \text{theta mean})}{\lambda_i * \sqrt{\text{theta variance}}}$$

Convert IRT to IFA:

$$\lambda_i = \frac{a_i}{\sqrt{\text{theta variance}}}$$

$$\tau_i = a_i b_i + \frac{a_i * \text{theta mean}}{\sqrt{\text{theta variance}}}$$

Note: These formulas
 rescale a_i and b_i so
 that $\text{theta M}=0$, $\text{VAR}=1$

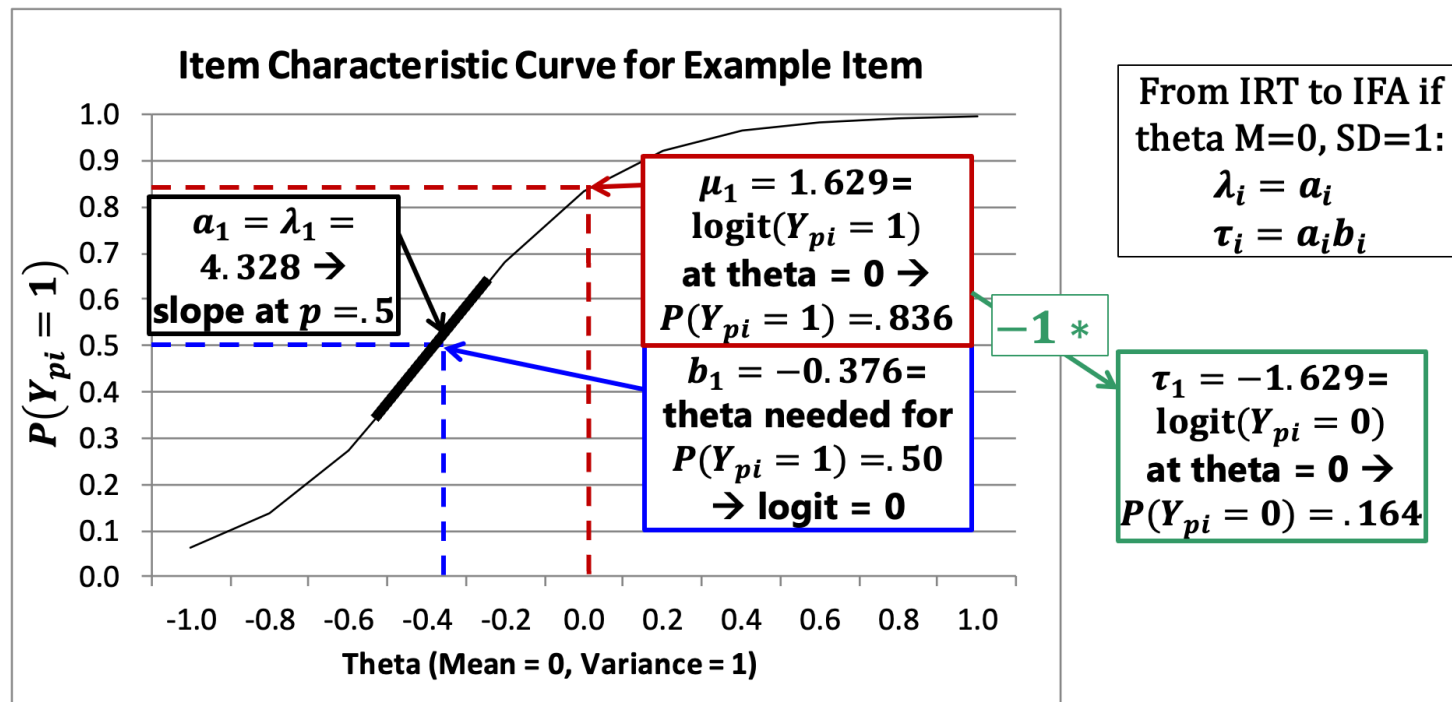
If you don't want to
 rescale theta, use $\text{M}=0$
 and $\text{VAR}=1$ for it to
 keep your current scale

Item Parameter Interpretations

IFA with loading and "easiness" **intercept** μ_i : $\text{logit}(Y_{pi} = 1) = \mu_i + \lambda_i \theta_p$

IFA with loading and "difficulty" **threshold** τ_i : $\text{logit}(Y_{pi} = 1) = -\tau_i + \lambda_i \theta_p$

2PL IRT with **discrimination** and **difficulty** b_i : $\text{logit}(Y_{pi} = 1) = a_i(\theta_p - b_i)$



Item Parameter Interpretations

IFA with loading and "easiness" **intercept** μ_i : $\text{logit}(Y_{pi} = 1) = \mu_i + \lambda_i \theta_p$

IFA with loading and "difficulty" **threshold** τ_i : $\text{logit}(Y_{pi} = 1) = -\tau_i + \lambda_i \theta_p$

2PL IRT with discrimination and **difficulty** b_i : $\text{logit}(Y_{pi} = 1) = a_i(\theta_p - b_i)$

- IFA and 2PL IRT item slope parameters are interpreted similarly:
 - IFA loading $\lambda_i = \Delta$ in logit of $Y_{pi} = 1$ per unit Δ in theta
 - IRT discrimination a_i = slope of ICC at prob=.50 (where logit = 0)
- IFA and 2PL IRT item location parameters are different:
 - **IFA intercept** μ_i = logit of $Y_{pi} = 1$ when **theta (x) = 0**
 - **IFA threshold** τ_i = logit of $Y_{pi} = 0$ when **theta (x) = 0**
 - **IRT difficulty** b_i = amount of theta needed for **logit (Y_{pi}) = 0**
 - So b_i difficulty values are more useful (to me) to index **location**

From IFA to IRT

- This yields the discrimination difficulty parameterization that is common in unidimensional IRT models:

$$P(Y_{pi} = 1 \mid \theta_p) = \frac{\exp(a_i (\theta_p - b_i))}{1 + \exp(a_i (\theta_p - b_i))}$$

- Here,
 - b_i is the item difficulty—the point on the θ scale at which a person has a 50% chance of answering with a one
 - a_i is the item discrimination—the slope of a line tangent to the curve at the item difficulty
- IRT models have a number of different forms of this equation (this is the two-parameter logistic 2PL model)

IRT Example: Achievement Data

To demonstrate a couple IRT models, we will compare a 1PL and 2PL model for the example data

- Please see the script `mlmmWorkshop2025_Lecture02_Introduction_To_Measurement_Models.R` for details

Generalized Linear (Psychometric) Models: Summary

- A key to understanding the varying types of psychometric models is that they must map the theory (the right-hand side of the equation— θ_p) to the type of observed data (left-hand side of the equation)
- So far we've seen two types of data: continuous (with a normal distribution) and binary (with a Bernoulli distribution)
- For each, the right-hand side of the item model was the same
- For the normal distribution:
 - We had an error term but did not transform the right-hand side
- For the Bernoulli distribution:
 - No error term and a function used to transform the right-hand side so that the conditional mean will range between zero and one